



ORIGINAL ARTICLE

Patterns of Herbivory and Resource Utilization of Tree Seedlings Along an Altitudinal Gradient

Jordon C. Tourville^{1,2*}  and Martin Dovciak¹

¹Department of Environmental Biology, State University of New York College of Environmental Science and Forestry, 1 Forestry Drive, Syracuse, New York 13210, USA; ²Research Department, Appalachian Mountain Club, PO Box 298, Gorham, New Hampshire 03818, USA

ABSTRACT

In montane areas, climate change can shift tree species distributions upslope over time which can affect forest ecosystem structure and functioning. Seedlings of low-elevation temperate broadleaf trees establishing beyond their ranges at high elevations need to overcome both herbivore pressure and soil nutrient limitations. To assess the influence of these two stressors, we quantified leaf and stem herbivory, soil and foliar nutrient status, and two-year survival of experimentally grown seedlings of two hardwood species, sugar maple (*Acer saccharum*) and American beech (*Fagus grandifolia*) along elevation gradients in the Green Mountains of Vermont, USA. While insect foliar herbivory was reduced on maple seedlings growing beyond range boundaries, suggesting enemy escape, the opposite pattern was observed for beech. Mammalian browsing increased with elevation for both species. In general, foliar nutrient

concentrations and resource utilization (that is, the relationship between soil nutrient availability and foliar nutrient status) declined with elevation for both species (although more so for maple, especially calcium), while toxic foliar aluminum concentrations increased for maple. Survival decreased with elevation for both species, but especially for sugar maple, linked more to declining foliar nutrient status than herbivory at higher elevations. Thus, the effects of herbivory and nutrient utilization on seedling survival can be critical for shaping tree species range shifts and, ultimately, species composition and forest ecosystem functioning.

Key words: Browse; climate change; herbivory; montane spruce-fir; northern hardwoods; nutrients; seedlings; soils

HIGHLIGHTS

- Patterns of herbivory vary for individual tree species across elevation
- Soil nutrient utilization tracks seedling survival across range boundaries
- Differential herbivory and resource utilization can help shape tree range shifts

Received 28 February 2025; accepted 15 July 2025;
published online 11 August 2025

*Corresponding author; e-mail: jtourville@outdoors.org

INTRODUCTION

Climate change has been driving observable shifts in species distributions on both local and global scales (Boonman and others 2024; Lawlor and others 2024). While conventional theories and some empirical data suggest that tree species adjust to climate by shifting their range limits toward higher elevations and latitudes, certain species have demonstrated unexpected shifts toward warmer climates, potentially influenced by historical disturbances and ongoing successional processes (Foster and D'Amato 2015; Wason and Dovciak 2017; Tourville and others 2022). Many trees, particularly deciduous species, may be particularly slow to migrate due to their slow growth, short dispersal distances, and long lifespans, factors that contribute to slow population turnover and a restricted capacity to adapt to swiftly changing environments (Aitken and others 2008). Additionally, some research, particularly in northeastern North America, indicates slow or delayed tree species range shifts compared to what might be expected given current warming trends, often attributed to soil conditions and belowground biotic interactions (Brown and Vellend 2014; Alexander and others 2018; Vellend and others 2021; Tourville and others 2023a, 2024). Despite numerous efforts to model and forecast future tree species distributions in the context of climate change, few studies have incorporated these edaphic abiotic or biotic variables (Bueno de Mesquita and others 2016).

For trees, range expansion requires the successful establishment and subsequent growth of seedlings (Tourville and others 2022; Chardon and others 2024). Seedling establishment, encompassing germination, emergence, and the initial years of survival, is particularly sensitive to both physical factors (such as temperature, moisture, and nutrients) and biotic factors (including litter depth, seed and seedling predators, and pathogens) (Leck and others 2008). Consequently, seedling establishment is considered more susceptible to climate change than adult tree growth and survival, potentially creating a bottleneck to population persistence and establishment at range edges and beyond (Walck and others 2011; Tourville and others 2022). Therefore, early life stages represent a crucial focus in trees species range shifts research.

Biotic interactions may impede or delay seedling establishment through various mechanisms, further complicated by global environmental changes (Alexander and others 2018). Plants are influenced by other biota both negatively, through antagonistic interactions with competitors, herbivores, or

pathogens, and positively, through mutualistic interactions with pollinators, seed dispersers, or mycorrhizal fungi (Brown and Vellend 2014; Maron and others 2014; Urli and others 2016; Crofts and Brown 2020; Evans and others 2020; Tourville and others 2023a, Tourville and others 2024). In general, herbivory on roots and leaves can hinder survival and growth by reducing a plant's resource acquisition ability (Crawley 2009), while herbivory on seeds and flowers (for example, by rodents and insects) can affect plant abundance by diminishing sexual reproduction (Maron and Crone 2006; Crofts and Brown 2020; Tourville and others 2023b). Selective mammalian browsing has been suggested to restrict or "cool" range expansions of some temperate tree species under changing climate (Fisichelli and others 2012). In general, while plant–animal interactions like pollination and herbivory have been shown to impact individual plant fitness and population growth (Maron and others 2014), their role in the context of climate-driven range shifts remains under-explored.

Tree species population expansion (range shifts) can lag behind changing climate in the presence of strong negative interactions (such as seed predation or competition) or absence of strong positive interactions (such as the absence of mycorrhizal mutualists) at or beyond their leading range edge (van der Putten 2012; Rivest and Vellend 2018; Tourville and others 2023b, 2024). Species colonizing new areas or occurring at low densities may experience "enemy release" (van der Putten 2012; Brian and Catford 2023), a mechanism stemming from the Janzen–Connell hypothesis, which suggests that plants in low-density patches benefit from a reduction in herbivore pressure. Other theories predict a resource dilution effect at high densities, leading to greater herbivory in lower density plant patches (Wetzel and others 2023). Variation in the magnitude of herbivory along gradients of elevation, canopy openness, and disturbance has been reported in many studies (Wetzel and others 2023). Among those conducted on range limits, some have found increases (Fisichelli and others 2012; Rivest and Vellend 2018; Crofts and Brown 2020; Evans and others 2020), while others have found decreases (Urli and others 2016) in herbivory toward plant species' range limits. More research is needed to allow for a better understanding of the variation in the strength of plant–animal interactions across species' range limits.

Montane slopes in northeastern North America are dominated by deciduous temperate (northern hardwood) tree species at lower elevations and

coniferous spruce-fir tree communities at higher elevations (for example, Wason and Dovciak 2017). In northern hardwood forests at lower elevations, deeper, well-developed, and relatively nutrient-rich soils typically transition to rocky, shallow, acidic, and nutrient-poor soils at higher elevations (Collin and others 2017; Evans and Brown 2017). Soil acidity in this region is influenced not only by variation along elevational gradients but also by disparities in parent material and anthropogenic acidification resulting from atmospheric nitrogen (N) and sulfur (S) deposition (Zarfos and others 2019; Benish and others 2022). Although N and S deposition has significantly decreased following the 1990 amendments to the Clean Air Act, soils across the Northeast have been acidified, leading to depletion of soil base cations and mobilization of toxic inorganic aluminum (Al) (Lawrence and others 2015). However, long-term consequences of atmospheric nutrient deposition and subsequent recovery will have their own important impacts on forest composition and functioning (Gilliam and others 2019; Gilliam and others 2024; Houle and others 2024; Phelan and others 2024).

Sugar maple (*Acer saccharum* Marshall) and American beech (*Fagus grandifolia* Ehrh.), two ecologically and economically significant deciduous tree species co-dominating at lower elevations, may migrate upslope with accelerating climate warming (Urli and others 2016); however, current evidence does not suggest any contemporary upslope range shifts (Tourville and others 2022). While sugar maple generally exhibits higher cold tolerance compared to American beech, its growth and survival are sensitive to low base cation and high aluminum concentrations typical of acidic soils (Schaberg and others 2006; Bal and others 2015; Halman and others 2015), potentially hindering its upslope migration. On the other hand, American beech has demonstrated greater tolerance to acidic, base-poor soils (Sullivan and others 2013; Tourville and others 2023a) and could potentially perform better than sugar maple at higher elevations. However, to better understand the patterns of tree species advancement across their range limits, one needs to consider how these species uptake, partition, and utilize available soil resources in foliar tissues (for example, sugar maple requires relatively more soil Ca than American beech; Schaberg and others 2006; Cleavitt and others 2011; Halman and others 2015). Given the differences in soil resource utilization, climate-induced changes to species distributions (and ultimately future forest composition) could also be modulated by edaphic

conditions, affecting both future forest ecosystem function and forest management practices.

Study Questions and Hypotheses

Our overarching goal was to assess how herbivory and soil resource utilization vary for sugar maple and American beech seedlings along co-occurring elevation and soil acidity gradients in the northeastern USA, and how they may affect seedling survival. Here, we define resource utilization as a standardized ratio between site-level soil nutrient status and foliar nutrient status—an indication of how tree seedlings use available soil resources (an illustration of resource strategies: acquisitive versus conservative; see Ci and others (2022)). We were particularly interested in (1) evidence of mammalian browsing and insect foliar herbivory both within and beyond the elevational range of sugar maple and American beech, (2) resource utilization of critical soil nutrients (including base cations) and metals (for example, Al) across sugar maple and American beech elevational range boundaries, and (3) the effect between levels of herbivory and resource availability on two-year seedling survival within and beyond elevational range boundaries. We hypothesized that both mammalian browsing and insect foliar herbivory would be reduced for both sugar maple and American beech seedlings beyond their range margins, consistent with an enemy-release effect (**H1**). We also hypothesized that seedling soil resource utilization would be higher within a species range than beyond (**H2**). Lastly, we hypothesized that seedling survival would be higher for individuals experiencing less herbivory and higher concentrations of foliar nutrients and resource utilization (**H3**).

METHODS

Study Area

All fieldwork was conducted across edaphic and climate gradients associated with elevation on four mountains in the Green Mountains of Vermont, USA (see Figure 1 and Supplemental Table S1). The Green Mountains are distinguished by pronounced zonation in forest communities. At lower elevations, temperate deciduous northern hardwood forests, predominantly consisting of sugar maple and American beech, prevail, while montane spruce-fir forests, characterized by balsam fir (*Abies balsamea* (L.) Mill.) and red spruce (*Picea rubens* Sarg.), dominate higher elevations (Siccama 1974). These forest communities are delineated by a distinct ecotone situated approximately 800 m above

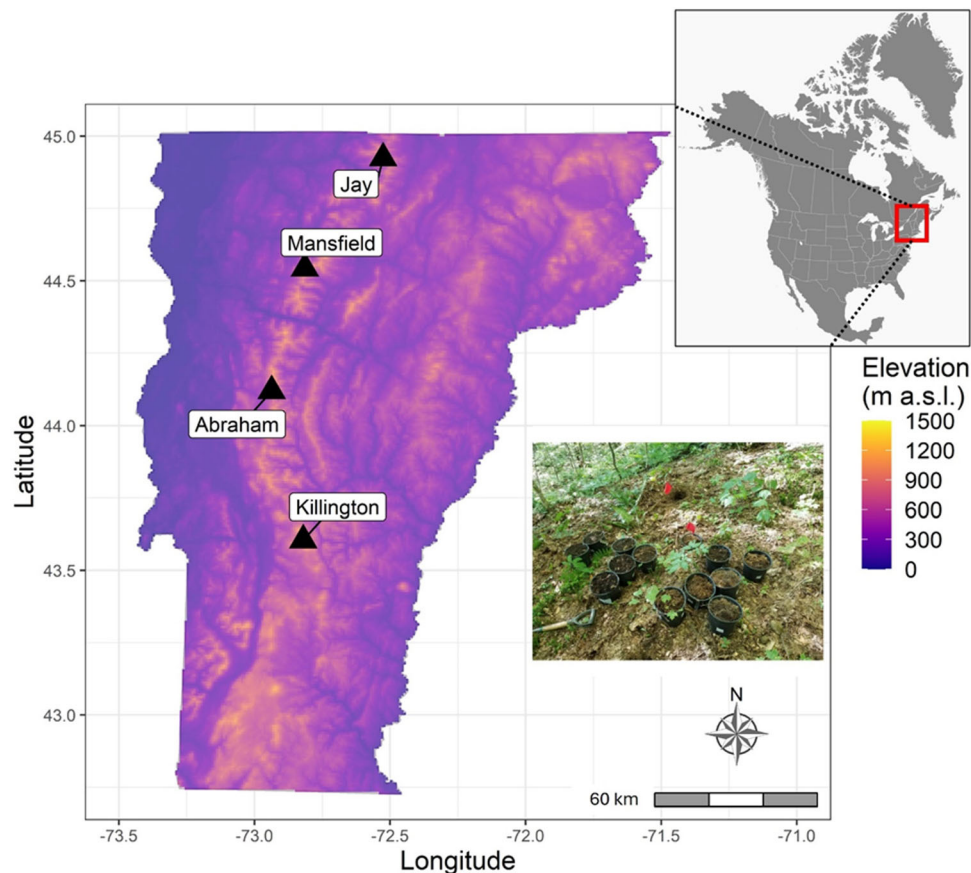


Figure 1. Location of the four study mountains in the Green Mountains of Vermont and the relative elevations (meters above sea level) within the landscape. The photograph on the right shows the experimental setup in the fall of 2019. Control seedlings were grown in black plastic pots marked with red flagging as seen in the image (map projection: WGS84; photograph credit: Jordon Tourville).

sea level (asl) (Wason and Dovciak 2017). The northern region of New York and New England is characterized by highly varied terrain, typically ranging from 150 to 1,220 m asl, rocky spodosol soils, and a continental climate marked by warm summers and cold snowy winters (with mean annual temperatures ranging between 3 and 11 °C, a mean winter minimum of -12.2 °C, a mean summer maximum of 27.2 °C, and a mean length of the frost-free period of approximately 100 days). The region experiences evenly distributed precipitation throughout the year, with an annual mean precipitation of 890 mm in Albany, New York (Janowiak and others 2018).

Study Design

To evaluate the elevational patterns of foliar nutrient status (and utilization) and herbivory, and their overall influence on seedling survival for sugar maple and American beech, data were obtained from within an existing large mycorrhizal field

study (see the details in Tourville and others (2023, 2024); and below; from “Experiment 2”). This experiment was conducted on 12 sites in total that were established on four mountains in Vermont in the northeastern USA (Mount Abraham, Mount Mansfield, Jay Peak, and Killington Peak) at three elevations on each mountain (at every 200 m in elevation; 600 m = northern hardwoods site, 800 m = ecotone site, 1000 m = spruce-fir site) (Figure 1, Supplemental Table S1).

For the present study, unmanipulated soils were carefully extracted and placed inside 3-L plastic pots (which themselves were placed in the locations where soils were extracted) with manually cut holes covered with a fine mesh (accounting for a total of 50% pot surface area) to allow water transport within each pot. Twenty seeds of either sugar maple or American beech were sown in each pot in October 2019, and we tracked germination and survival for two growing seasons (age of all final surviving seedlings is 2 years). Two pots per

species were placed next to one another forming a single experimental block, and six blocks per site (elevation) were established, all at least 15 m apart (2 pots per species $\times$ 6 blocks $\times$ 3 sites $\times$ 4 mountains = 144 total pots per species).

All seeds were purchased from Sheffield Seed Company (Locke, NY), which source seeds from populations within the same region as this study, surface-sterilized with 2% H_2O_2 for 2 h, and cold-stratified under field conditions to break dormancy. The top of each pot was covered with a 0.5-cm plastic mesh to protect against seed predation during the winter months. The mesh was removed after snowmelt the following year to allow seedling growth. Overall germination rates ranged between 20 and 40% in each pot (max per pot = 7, median = 3, 864 total seedlings germinated, see Tourville and others (2024)). Final average experimental seedling densities for both sugar maple and American beech across all sites were 0.8 per m^2 , while natural seedling background densities were 3.1, 1.9, and 0 per m^2 averaged between both species for hardwood sites, ecotone sites, and spruce-fir sites, respectively (Tourville and others 2022).

Site visits occurred twice a month starting in May 2020. Seedling survival and seedling height was recorded for each individual seedling, while the presence/absence of browsing (stem damage) or foliar herbivory was recorded for each individual pot during each visit (final 2-year seedling heights ranged between 3 and 21 cm). Specifically, stem damage as a result of obvious mammalian browsing (bark striping or straight angled cuts) was recorded as present or absent. The stem herbivory we observed was characteristic of ungulate browsing; jagged and torn straight cut on the remaining stems were likely due to white-tailed deer (*Odocoileus virginianus*) or moose (*Alces alces*) or a clean 45° cut for snowshoe hares (*Lepus americanus*) (Williams and others 2000). Only foliar damage as a result of obvious insect herbivory (uniform holes, leaf mining, tattered leaves) was recorded as present/absent following Urli and others (2016). Final seedling harvests were done in September 2021 following two full growing seasons. All above-ground material was separated from seedling roots, dried at 60 °C for 24 h, and weighed.

Soil and Foliar Sampling

To assess the native soil chemistry of each site (see Tourville and others 2022, 2023), a total of 30 soil cores (5 cm diameter $\times$ 15 cm depth, outside of the pots) were extracted from each site surround-

ing the pots in the fall of 2019. Five cores were mixed to form a composite sample ultimately yielding 6 soil samples per site (3 sites $\times$ 4 mountains = 72 samples total). All soil cores were processed to separate the mineral upper B horizon from the organic O horizon and sieved using a 2-mm sieve to remove large debris. Sieved upper B horizon soils were processed to quantify soil pH via electrometric methods, and soil organic matter (SOM) was determined via the loss on ignition procedure in the Soil Analysis Laboratory at SUNY-ESF. Additionally, soil subsamples from each elevation were sent to the University of Vermont Agricultural and Environmental Testing Lab to analyze for effective cation exchange capacity (ECEC, sum of exchangeable base cations plus exchangeable acidity), base cation concentrations (Ca, Mg, K, Na), and metal concentrations (Al) using a modified Morgan procedure (McIntosh 1969). Due to logistical constraints, soil nitrogen (NH_4^+ or NO_3^-) was not analyzed as part of this study. Leaves from each surviving seedling were pooled by species and site following drying, finely ground using a Wiley mill, and shipped to the Penn State Agricultural Analytical Services Laboratory for tissue analyses. Total nitrogen, phosphorus, potassium, calcium, and other trace elements were extracted via acid digestion and ICP analysis (Huang and Schulte 1985).

Statistical Analyses

Hypothesis 1 We constructed generalized linear mixed models (GLMMs) to analyze the response of observed seedling foliar herbivory and browse (presence/absence per pot assuming a binomial distribution of residuals) to planting site (elevation) and using study mountain as a random effect. The soil chemical gradient on all study mountains was characterized with one-way ANOVA and post hoc Tukey's HSD tests (pH, soil organic matter, cation exchange capacity, and concentrations of Al, P, K, Ca, and Mg). We used linear mixed models (LMMs) to evaluate differences in foliar nutrient concentrations (N, P, K, Ca, Mg, Al) for sugar maple and American beech seedlings across planting sites. Raw data were averaged across all seedlings within each replicate. Model assumptions and standard diagnostics were checked for each GLMM and LMM.

Hypothesis 2 We constructed a correlation matrix using Spearman's rank correlation coefficient for pairwise comparisons of site-level soil chemical variables and equivalent nutrient status (see be-

low), and site and species-level foliar nutrient status (note that soil N was not measured in this study). We calculated a site-level unitless index value for resource utilization we called the Normalized Relative Utilization Score (NRUS). This was calculated for each nutrient (P, K, Ca, Mg, Al) by scaling and centering foliar values and their soil equivalents, then dividing by each other, and subtracting by the site-level soil mean to center the score. This results in an interpretable and relative score of utilization for each nutrient at each site within a species. Larger scores indicate that seedling leaves contain a relatively greater share of a particular nutrient given the amount available in the soil at that site, while smaller scores indicate the opposite.

Hypothesis 3 GLMMs were used to analyze the response of survival (binomial) to planting site (elevation), foliar herbivory, and browsing (and their interactions) averaged for individuals within the same block and using study mountain as a random effect. Survival was standardized by the number of successful germinants in each pot at the beginning of the experiment. Models for seedling survival with scaled and centered individual foliar nutrient concentrations as predictors were also constructed, with interactions with elevation examined as well. We also constructed GLMMs for all the same fixed effects with dried aboveground biomass as a response, and coefficient estimates are presented in Supplemental Figure S1. All of the above analyses were conducted in R using the following packages: “dplyr,” “reshape2,” “ggplot2,” “Results” and “sjPlot” (R Core Team 2024).

Patterns of Herbivory

For sugar maple, the observed proportion (%) of seedling pots browsed was greater in the ecotone (30%) and above species range limits (spruce-fir forest; 38%) than within temperate hardwood forests (9%). For American beech, observed browse was also higher in spruce-fir (10% of pots) forests than within its range ($\leq 5\%$ of pots) ($p < 0.05$ for both species, not supporting **H1**; Figure 2A, Table 1). Observed foliar herbivory was higher for sugar maple seedlings in hardwood forests (85% of pots) and the ecotone (73% of pots) compared to spruce-fir forests (51% of pots) ($p \leq 0.02$, partially supporting **H1**), although this pattern was reversed for beech seedlings, with less foliar damage observed in hardwood (35% of pots) forests compared to ecotone (59% of pots) or spruce-fir (47% of

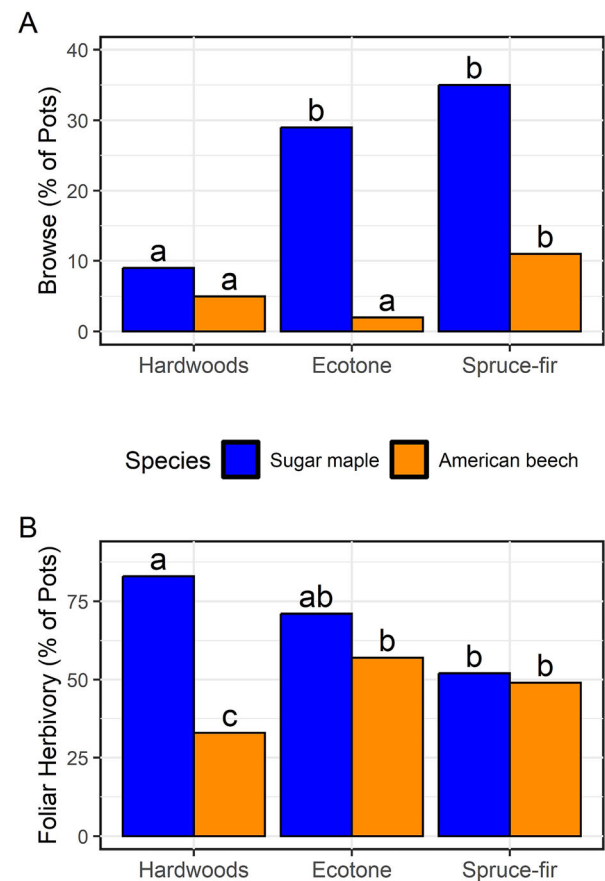


Figure 2. **A** Proportion of pots with observed browsing and **B** foliar herbivory on sugar maple (blue) and beech (orange) seedlings over the course of two growing seasons at three different elevations. Letters indicate significant differences between groups from GLMMs (planting sites, $\alpha = 0.05$).

pots) forests ($p = 0.04$, not supporting **H1**, Figure 2B, Table 1).

Soil and Foliar Chemistry and Resource Utilization

Within the upper B soil horizon, pH and ECEC averaged across all study mountains tended to decrease at higher elevations, while SOM increased (Figure 3A, Supplemental Table S2). Liable P and Mg concentrations peaked at ecotone sites, while Al concentrations peaked at spruce-fir sites, although Mg concentrations were not significantly different between sites (Figure 3A, Supplemental Table S2). K concentrations tended to increase, but not significantly with elevation, while Ca concentrations tended to decrease with increasing elevation (Figure 3A, Supplemental Table S2). Soil Ca:Al declined steeply with elevation: hardwoods = 3.48;

Table 1. Model Summaries of Planting Site (Elevation), Browse, Foliar Herbivory, and Individual Foliar Nutrient Effects on Sugar Maple and American Beech Seedling Survival, Including Browse and Foliar Herbivory $\times$ Planting Site Interactions (L = Hardwoods, M = Ecotone, H = Spruce-fir)

Response	Factor	Sugar maple				American beech			
		Estimate	SE	t-value	Prob	Estimate	SE	t-value	Prob
Browse	Intercept	− 0.57	0.30	− 1.96	0.05*	− 2.08	0.78	− 2.65	0.008**
	Planting Site Hardwoods	− 1.84	0.61	− 3.03	0.002**	− 1.25	0.64	− 2.49	0.01**
	Ecotone	− 0.32	0.44	− 0.73	0.47	− 1.98	0.89	− 2.80	0.007**
Herbivory	Intercept	0.99	0.52	1.96	0.05*	0.02	0.63	0.04	0.97
	Planting Site Hardwoods	2.34	0.59	3.94	< 0.001***	− 0.83	0.45	− 2.81	0.04*
	Ecotone	1.07	0.47	2.26	0.02*	0.33	0.46	0.91	0.47
Survival	Intercept	0.21	0.05	4.09	< 0.001***	0.09	0.09	1.04	0.35
	Planting Site Hardwoods	0.14	0.06	2.27	0.02*	0.27	0.06	4.42	< 0.001***
	Ecotone	− 0.05	0.07	− 0.83	0.41	− 0.03	0.04	− 0.43	0.64
Survival	Browse	− 0.19	0.08	− 5.65	< 0.001***	− 0.09	0.11	− 0.76	0.25
	Browse x Planting Site: L-H	0.01	0.05	0.13	0.98	0.08	0.28	0.29	0.91
	Browse x Planting Site: M-H	− 0.02	0.06	− 0.11	0.99	− 0.09	0.28	− 0.31	0.93
	Herbivory	− 0.18	0.06	− 2.59	0.01**	− 0.07	0.07	− 0.19	0.78
	Herbivory x Planting Site: L-H	0.24	0.08	2.83	0.007**	− 0.27	0.25	− 0.79	0.31
	Herbivory x Planting Site: M-H	0.06	0.06	0.71	0.14	− 0.03	0.16	− 0.19	0.87
	N	0.10	0.05	2.73	0.01**	0.13	0.10	2.75	0.01**
	P	0.01	0.03	0.51	0.62	0.02	0.03	0.79	0.40
	Ca	0.22	0.09	4.31	< 0.001***	0.05	0.06	0.91	0.37
	K	0.23	0.09	4.15	< 0.001***	0.14	0.07	4.01	< 0.001***
	Mg	0.20	0.08	3.98	< 0.001***	0.15	0.07	5.23	< 0.001***
	Al	− 0.43	0.13	− 4.56	< 0.001***	− 0.04	0.05	− 1.67	0.12

Note that planting site is treated as categorical in this table and continuous in coefficient figure, also showing model summary of planting site (elevation) effects on observed sugar maple and American beech seedling browse and foliar herbivory probabilities. Results are from generalized linear mixed models assuming a binomial distribution of residuals for survival, browse, and foliar herbivory. Mountain was held as a random effect. Probabilities in bold indicate $p \leq 0.05$. *($p \leq 0.05$), **($p \leq 0.01$), ***($p \leq 0.001$).

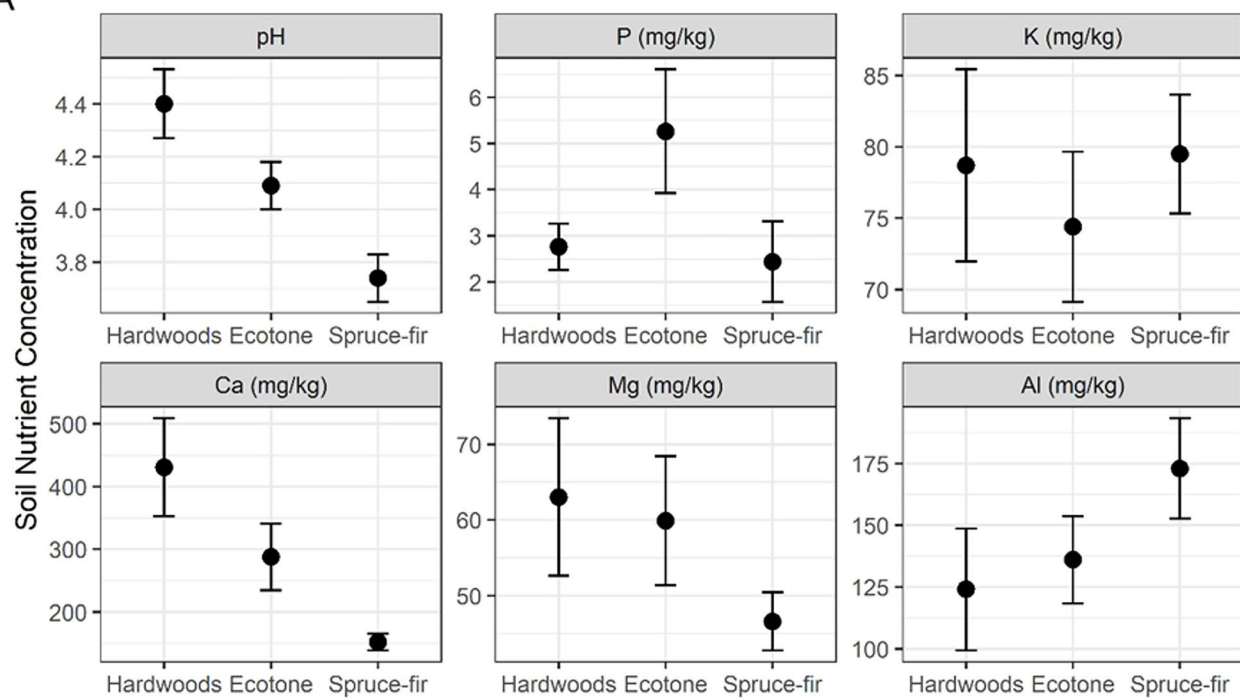
ecotone = 2.12; spruce-fir = 0.88 (Supplemental Table S2).

Across the elevation gradient, sugar maple foliar N and base cation (K, Ca, Mg) concentrations tended to decrease with increasing elevation (Figure 3B, Supplemental Table S3). Sugar maple Al foliar concentrations increased with increasing elevation and foliar P peaked within ecotone forests in concert with soil P. For American beech seed-

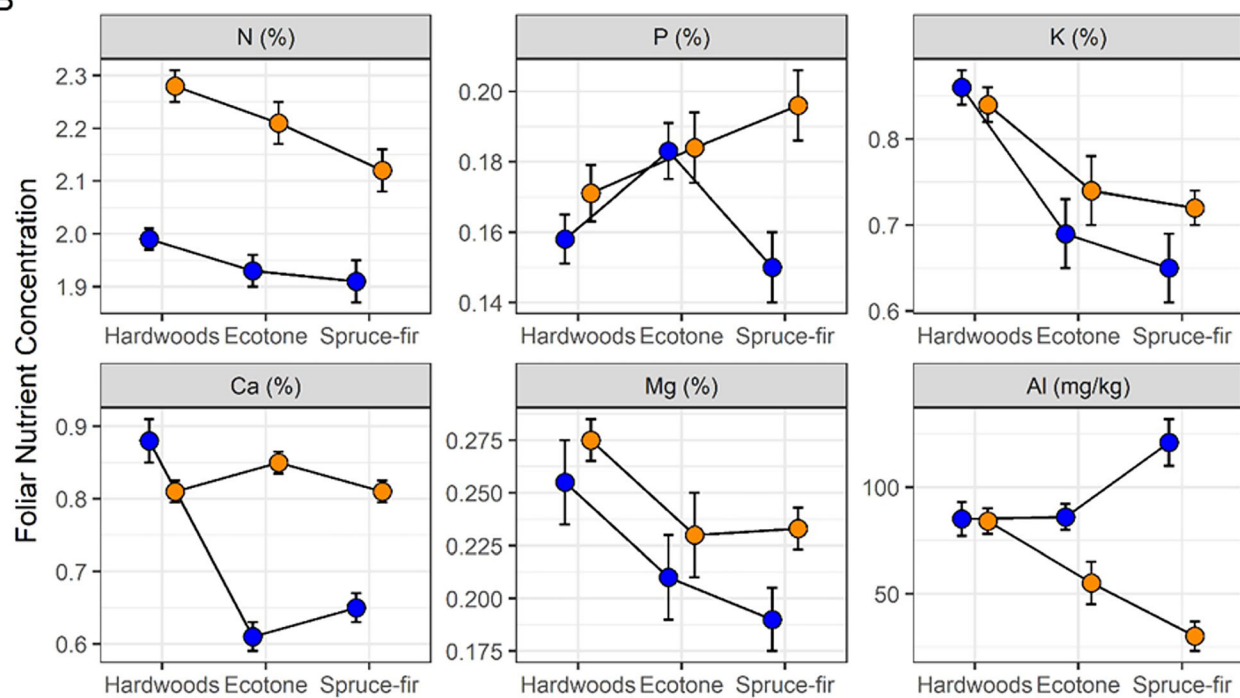
lings, we found a positive relationship between foliar P concentration and elevation and a negative relationship between foliar N, K, Mg, and Al concentrations and elevation (Figure 3B, Supplemental Table S3). No relationship was detected between foliar Ca concentration and elevation in beech.

The strongest observed correlation between most foliar and soil chemical variables was with soil SOM (negative) and soil pH (positive) for both seedling

A



B



Species ● Sugar maple ● American beech

◀ **Figure 3.** **A** Raw soil nutrient and chemical value (pH) means ($\pm$ SE) across three sites/elevations on four mountains in the Green Mountains of Vermont. **B** Raw foliar nutrient mean ($\pm$ SE) values for harvested sugar maple (blue) and American beech (orange) seedlings from our planting experiment partitioned across three different elevations (low elevation = Hardwoods, mid-elevation = Ecotone, high elevation = Spruce-fir).

species (Supplemental Figure S2). Soil and foliar chemical equivalents mainly displayed weak correlations ($-0.2 > r < 0.2$), the exception being P for both species ($-0.3 < r > 0.3$) and Al for sugar maple ($r = 0.4$). NRUS of sugar maple for Al was greatest at high-elevation spruce-fir sites, and for P, Ca, K, and Mg, greatest at low-elevation hardwood sites (supporting **H2**; Figure 4A and B). NRUS of American beech for P, Ca, and Mg was highest at high-elevation spruce-fir sites, and for K and Al highest at low-elevation hardwood sites (partly supporting **H2**; Figure 4A and B).

Impacts on Seedling Survival

Survival of both sugar maple and American beech decreased beyond their range margins (sugar maple: hardwood = 81% survival vs. spruce-fir = 19% survival: $p \leq 0.02$; beech: hardwood = 34% survival vs. spruce-fir = 17% survival: $p < 0.001$; Table 1). Browsing and foliar herbivory both significantly decreased the probability of survival for sugar maple, while foliar herbivory decreased beech survival but not significantly (partially supporting **H3**; Figure 5, Table 1). There was a significant interaction effect between foliar herbivory and elevation on survival, with greater mortality

related to herbivory at higher elevations ($p = 0.007$, Figure 5, Table 1), although this was not the case for browse $\times$ elevation interaction ($p = 0.98$). No significant interaction terms between browse, foliar herbivory, or any foliar nutrients and elevation were recorded for American beech. Further, sugar maple survival probability was significantly greater when site-level mean concentrations of all measured foliar nutrients were higher, with the exception of P, whereas beech survival was only significantly improved with higher site-level mean foliar concentrations of Mg and K (partially supporting **H3**; Figure 5, Table 1). Lastly, higher foliar concentrations of Al significantly decreased sugar maple survival ($p < 0.001$), while no pattern was observed for beech ($p = 0.12$, Figure 5, Table 1).

DISCUSSION

We find that the strength of antagonistic biotic interactions (browse and herbivory) can differ significantly across the elevational range of a plant species, from the core to its edge and beyond, but that individual species can display idiosyncratic patterns. Following **H1**, insect foliar damage was detected for sugar maple on the majority of experimental pots within the core of its range and then declined steadily with elevation, both at its range edge and beyond, consistent with an enemy-release effect (compare Janzen-Connell hypothesis; Janzen 1970; Connell 1971; Comita and others 2014). However, beech displayed the opposite patterns of foliar herbivory with elevation. Mammalian stem browsing increased for both seedling species with elevation, inconsistent with predictions from enemy release, but consistent with the

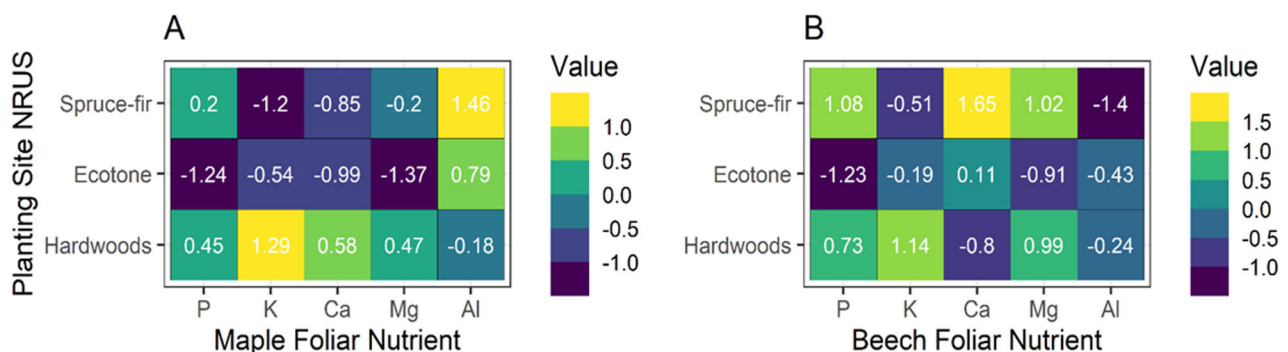


Figure 4. Normalized resource utilization score (NRUS) for each measured foliar nutrient at each site/elevation for both sugar maple (**A**) and American beech (**B**) seedlings. NRUS is calculated by dividing the scaled and centered foliar value by its soil equivalent and subtracting by the site-level soil mean to center the score. Larger scores indicate that seedling leaves contain a relatively greater share of a particular nutrient given the amount available in the soil at that site, while smaller scores indicate the opposite.

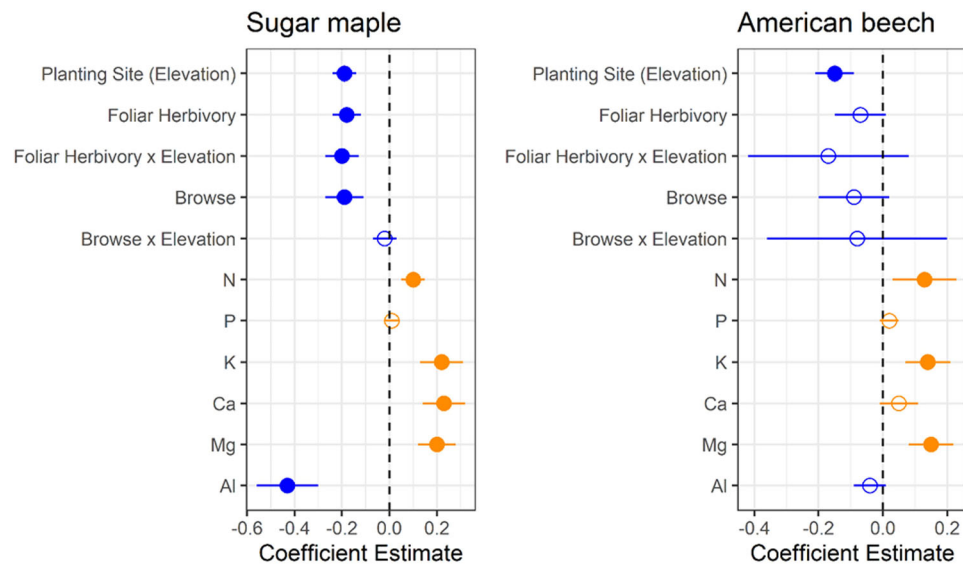


Figure 5. Fixed effect and interaction coefficient estimates from GLMM for sugar maple (left) and American beech (right) seedling survival (binomial) as a response and study mountain held as a random effect. Positive estimates are in orange and negative estimates are in blue. Filled points indicate estimates with standard errors not overlapping zero, and hollow points indicate estimates with standard errors overlapping zero. Nutrient values represent foliar nutrient concentrations. Note elevation (planting site) is treated as continuous here. Interaction coefficients (all non-significant) for foliar nutrients and elevation on seedling survival are not shown here.

idea of greater browsing pressure on temperate deciduous tree species establishing above their range limits (Fischelli and others 2012; Frelich and others 2012). Biotic interactions at and beyond range limits have been discussed elsewhere in the literature (HilleRisLambers and others 2013; Early and Keith 2019; Tourville and others 2024); however, few empirical studies have directly quantified the importance of multiple biotic interactions along natural environmental gradients. Given the disparity in the strength of different sources of herbivory across elevation, and further complicated by species-specific responses, the overall additive effect on plant performance may be hard to forecast.

Greater herbivore abundance is theorized to occur in less abiotically stressful conditions (Hargreaves, 2024), and many studies have detected increased herbivory at lower rather than upper elevation range limits (for example, by ungulate and aphids; Galen 1990; Goodwin 2025). Indeed, in the high-elevation montane conifer forests of the Northeast, deer and moose, the species most likely browsing our experimental seedlings, experience low population densities (Koitzsch 2002; Bergeron and others 2011). Given the high browse observed for both hardwood species, we posit that ungulates are displaying increased selectivity for particular seedling species in the spruce-fir forest. While hardwood seedling density is relatively low near

montane ecotones, the evidence of increased proportional herbivory by mammals at these sites could suggest increased palatability of these species compared to conifers. In another study looking at sugar maple performance beyond elevation range boundaries, Brown and Vellend (2014), using a granivore exclusion experiment, found that most seeds sown just beyond their range limit were consumed by seed predators, leading to very low regeneration. Tourville and others (2023b) used a seed choice experiment to assess seed species preference by native granivores and arrived at similar results for granivore preference. This suggests that both preferential herbivory and seed predation may create lags in upward elevational range expansions for northern hardwood species.

Focusing on **H2**, we find good evidence that soil resources are partitioned and utilized in foliar tissue the most efficiently within the ranges of our focal tree species. While positive correlations between sampled soil nutrient and foliar nutrient status were weak, patterns of foliar nutrients and foliar resource utilization all generally mirrored the elevation edaphic gradient uncovered from our soil nutrient sampling. High-elevation montane forests in the Northeast, based on this and previous studies, have low concentrations of base cations, low pH, and high mobilized inorganic aluminum and iron (Cleavitt and others 2011; Lawrence and

others 2015; Tourville and others 2023; Ni and Vellend 2024). Calculating nutrient utilization, by adjusting raw foliar nutrient concentrations by the availability of that nutrient in the soil, provided a cleaner picture of how plants are responding to the edaphic gradient. However, we acknowledge that given the importance of plant available N to seedling nutrition in our system, the lack of data on soil N is a drawback of our study. Still, based on utilization scores and previous research, it appears that sugar maple seedlings are more sensitive to poor (low pH and base cations) boreal soils than American beech (Sullivan and others 2013; Bal and others 2015; Lawrence and others 2015). The extremely high concentrations of foliar Al and NRUS of Al in sugar maple within spruce-fir sites are particularly striking. Al has been shown to dramatically increase sugar maple mortality via suppression of growth, and its high assimilation into foliar tissues at high elevations may be severely limiting to long-term successful establishment (Schaberg and others 2006; Halman and others 2015). It has also been shown to be tightly coupled with Ca depletion in the soil (via acidification), which is a critical nutrient for sugar maple (Schaberg and others 2006), and a nutrient with relatively low utilization at high elevations. Indeed, an examination of soil Ca:Al from our sampling bears this out (see Supplemental Table S2). Overall, we posit that the patterns of soil nutrient utilization by tree seedlings across elevation gradients, particularly as they related to Ca and Al utilization, can dictate the levels of seedling mortality observed in this system.

Recent work investigating factors that control seedling distributions indicates that both climate and soils are significant drivers (Tourville and others 2022). Assuming climate continues to become more favorable for both sugar maple and American beech in boreal forests at high elevations, acidic soils with low base cation saturation and high available Al may impede sugar maple seedlings but allow for the establishment of acid-tolerant beech seedlings and possibly other less abundant species such as red maple (*Acer rubrum*), striped maple (*Acer pensylvanicum*), mountain maple (*Acer spicatum*), and birch species (*Betula sp.*) (Bose and others 2017; Tourville and others 2022). Given the observed ability of American beech (via advanced seedling recruitment and vegetative sprouting) to compete with and suppress sugar maple seedlings (Nyland and others 2006), especially given the increase of sapling basal area and stem density largely as a result of beech bark disease, the establishment of beech at higher eleva-

tions may further hinder upslope sugar maple migration (cf. Cleavitt and others 2018). This apparent difference in soil chemical niche space is also supported by more recent work in the region (Ni and Vellend) and may have significant consequences for future forest composition both within the northern hardwood forest biome and at higher elevations where range expansions of these species may occur (Cleavitt and others 2011; Tourville and others 2022).

Our findings indicate that both the availability of soil nutrients to tree seedlings and the antagonistic biotic interactions they face are paramount for their long-term survival, following **H3**. For both focal species, but especially for sugar maple seedlings, foliar base cation nutrient concentrations and herbivory from browsing and feeding on foliar tissue are tied to patterns of survival and mortality (Cleavitt and others 2011; Fernando and others 2016). In general, seedling mortality was higher at higher elevations, areas with cooler temperatures (see Supplemental Table S1) lower nutrient utilization scores, lower base cation concentrations, higher aluminum concentrations, and greater mammalian browsing. However, while insect foliar damage was greater at lower elevation sites, sugar maple mortality was greater at higher elevations, despite the significant negative relationship between foliar herbivory and survival, indicating a strong herbivory $\times$ elevation interaction. Our experiment illustrated that both insect and mammalian herbivory was one potential cause of seedling mortality within the core of sugar maple's elevational range, with more than 70% of seedlings impacted by insect herbivores and $\sim 25\%$ impacted by mammal browsing. Despite this, patterns of beech survival and mortality were largely decoupled from herbivore activity. It should be noted that because the foliar nutrient status of our experimental seedlings was not compared with that of naturally occurring seedlings, differences in overall herbivory may not be fully reflective of the natural setting. However, given the local seed source of our seedlings and the direct comparison made only between experimental seedlings, we feel confident that our results here are suggestive of real-world dynamics.

Interspecific differences in herbivory intensity, with both browse and herbivory typically lower for beech and subsequently having little effect on its survival, are especially striking in light of the higher nutrient concentrations in beech leaves across the elevation gradient and the proximity of beech and maple seedlings within sites. On the surface, this would suggest that the less palat-

able tissues of sugar maple are preferentially targeted. However, we note that individual beech likely have higher constitutive defenses than sugar maple, such as a greater share of complex carbohydrates like lignin or polyphenols like tannins (Hane and others 2003) and possibly smaller leaf mass area (Fahey and others 2025). While we lack data on foliar carbon (C) or C:N, this could potentially be higher in beech than maple (Wills and others 2023), although further sampling would be required to flush out these possibilities. Regardless, we argue that patterns of herbivory also serve as a good indicator of forage quality among our seedling species.

The lag between the velocity of climatic change and velocity of tree range shifts may be caused by the biotic interactions and edaphic conditions we examined which together act to impede seedling establishment through increased mortality (see Alexander and others 2018). In some studies, this inhibition of tree range changes has been referred to as a “cooling” effect (Fisichelli and others 2012). Differential seedling establishment success between sugar maple and beech may have profound implications for the future recruitment of regionally dominant tree species and for future forest composition. Current forest tree species distribution models, such as landscape simulation models, are beginning to incorporate information on plant–animal interactions, disturbance regimes, and edaphic variables to better estimate the potential of species to keep pace with climate change (Bueno de Mesquita and others 2016; De Jager and others 2017), yet much more effort is required to parameterize and refine such models (Rammer and others ; Thom and others). The success of forest modeling and management efforts will depend largely on the ability to recognize and appreciate new establishment filters, especially in an era of increasing global change.

CONCLUSIONS

Our study, along with other recent studies (for example, Brown and Vellend 2014; Urli and others 2016; Rivest and Vellend 2018; Tourville and others), suggests that biotic interactions in tandem with abiotic factors, such as climate and soils, require greater consideration in order to provide reliable predictions of the consequences of climate warming for species distributions, although results are clearly species specific. Our study presents one of the few compelling case studies illustrating the alleviation of pressure from natural enemies at a leading range edge, albeit with the nuance that this

is only apparent for particular species (for example, sugar maple) and for one type of herbivory (that is, insect defoliation). Still, there is clear evidence indicating the central role of biotic interactions and soil chemistry in affecting species distribution shifts in response to climate change. However, forecasting these dynamics will be complicated by interactive and additive effects of these two drivers. Long-term demographic studies are needed to further test hypotheses within this realm. The overall impact of various biotic interactions can vary significantly between sites at the range edge and those immediately beyond it, underscoring the importance of geographically specific and detailed investigations of such interactions in anticipating species responses to global environmental changes. Ultimately, these and other results will contribute to the development of more sophisticated species distribution models that can be used to evaluate tree community shifts and inform the conservation and management of forest ecosystems in a changing world.

ACKNOWLEDGEMENTS

Research in the Green Mountains was supported partly by the US National Science Foundation award to Martin Dovciak (NSF award # 1759724) and partly by a Botanical Society of America Graduate Student Research Award, a New York Flora Association Research Award, and the Lowe-Wilcox Award from SUNY-ESF to Jordon Tourville. We thank Hailey Lynch, Heather Zimba, and Ben VanderStouw for their help with data collection; local, state, and federal organizations located in Vermont for providing research permits, access to sites, and logistic support; and the Forest Ecosystem Monitoring Cooperative and the Green Mountain Club for providing site access and logistic support. We thank Dr. Nathan Kiel for providing valuable feedback to early versions of this manuscript.

FUNDING

Research in the Green Mountains was supported partly by the US National Science Foundation award to Martin Dovciak (NSF award # 1759724) and partly by a Botanical Society of America Graduate Student Research Award, a New York Flora Association Research Award, and the Lowe-Wilcox Award from SUNY-ESF to Jordon Tourville.

DATA AVAILABILITY

The datasets and code generated during and/or analyzed during the current study are available for

download in GitHub (jtourvi, 2025) and are available in Zenodo at <https://doi.org/10.5281/zenodo.14940568>

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

REFERENCES

- Aitken SN, Yeaman S, Holliday JA, Wang T, Curtis-McLane S. 2008. Adaptation, migration or extirpation: climate change outcomes for tree populations. *Evolut Appl* 1(1):95–111. <https://doi.org/10.1111/j.1752-4571.2007.00013.x>.
- Alexander JM, Chalmandrier L, Lenoir J, Burgess TI, Essl F, Haider S, Kueffer C, McDougall K, Milbau A, Nuñez MA, Pauchard A, Rabitsch W, Rew LJ, Sanders NJ, Pellissier L. 2018. Lags in the response of mountain plant communities to climate change. *Glob Change Biol* 24(2):563–579. <https://doi.org/10.1111/gcb.13976>.
- Bal TL, Storer AJ, Jurgensen MF, Doskey PV, Amacher MC. 2015. Nutrient stress predisposes and contributes to sugar maple dieback across its northern range: a review. *Forestry* 88(1):64–83. <https://doi.org/10.1093/forestry/cpu051>.
- Benish SE, Bash JO, Foley KM, Appel KW, Hogrefe C, Gilliam R, Pouliot G. 2022. Long-term regional trends of nitrogen and sulfur deposition in the United States from 2002 to 2017. *Atmos Chem Phys Discuss* 2022:1–27.
- Bergeron DH, Pekins PJ, Jones HF, Leak WB. 2011. Moose browsing and forest regeneration: a case study in northern New Hampshire. *Alces* 47:39–51.
- Boonman CC, Serra-Diaz JM, Hoeks S, Guo WY, Enquist BJ, Maitner B, Svenning JC. . More than 17,000 tree species are at risk from rapid global change. *Nat Commun* 15(1):166.
- Bose AK, Weiskittel A, Wagner RG. 2017. A three decade assessment of climate-associated changes in forest composition across the north-eastern USA. *J Appl Ecol* 54(6):1592–1604. <https://doi.org/10.1111/1365-2664.12917>.
- Brian JI, Catford JA. 2023. A mechanistic framework of enemy release. *Ecol Lett* 26(12):2147–2166. <https://doi.org/10.1111/ele.14329>.
- Brown CD, Vellend M. 2014. Non-climatic constraints on upper elevational plant range expansion under climate change. *Proc R Soc B Biol Sci* 281:1794. <https://doi.org/10.1098/rspb.2014.1779>.
- Bueno de Mesquita CP, King AJ, Schmidt SK, Farrer EC, Suding KN. 2016. Incorporating biotic factors in species distribution modeling: are interactions with soil microbes important? *Ecography* 39(10):970–980. <https://doi.org/10.1111/ecog.01797>.
- Chardon NI, McBurnie L, Goodwin KJ, Pradhan K, Hille Ris Lambers J, Angert L. 2024. Variable species establishment in response to microhabitat indicates different likelihoods of climate-driven range shifts. *Ecography*. <https://doi.org/10.1111/ecog.07144>.
- Ci H, Guo C, Tuo B, Zheng LT, Xu MS, Sai BL, Cornelissen JH. 2022. Tree species with conservative foliar nutrient status and strong phosphorus homeostasis are regionally abundant in subtropical forests. *J Ecol* 110(7):1497–1507. <https://doi.org/10.1111/1365-2745.13884>.
- Cleavitt NL, Fahey TJ, Battles JJ. 2011. Regeneration ecology of sugar maple (*Acer saccharum*): Seedling survival in relation to nutrition, site factors, and damage by insects and pathogens. *Can J For Res* 41(2):235–244. <https://doi.org/10.1139/X10-210>.
- Cleavitt NL, Battles JJ, Johnson CE, Fahey TJ. 2018. Long-term decline of sugar maple following forest harvest, Hubbard Brook experimental forest, new hampshire. *Can J For Res* 48(1):23–31. <https://doi.org/10.1139/cjfr-2017-0233>.
- Collin A, Messier C, Bélanger N. 2017. Conifer presence may negatively affect sugar maple's ability to migrate into the boreal forest through reduced foliar nutritional status. *Ecosystems* 20:701–716. <https://doi.org/10.1007/s10021-016-0045-4>.
- Comita LS, Queenborough SA, Murphy SJ, Eck JL, Xu K, Krishnadas M, Zhu Y. 2014. Testing predictions of the Janzen–Connell hypothesis: a meta-analysis of experimental evidence for distance-and density-dependent seed and seedling survival. *J Ecol* 102(4):845–856. <https://doi.org/10.1111/1365-2745.12232>.
- Connell JH. 1971. On the role of natural enemies in preventing competitive exclusion in some marine animals and in rain forest trees. *Dyn Popul* 298(312):1–610.
- Crawley MJ. 2009. Natural enemies: the population biology of predators, parasites and diseases. Hoboken: John Wiley and Sons.
- Crofts AL, Brown CD. 2020. The importance of biotic filtering on boreal conifer recruitment at alpine treeline. *Ecography* 43:914–929. <https://doi.org/10.1111/ecog.04899>.
- De Jager, N. R., Drohan, P. J., Miranda, B. M., Sturtevant, B. R., Stout, S. L., Royo, A. A., & Romanski, M. C. (2017). Simulating ungulate herbivory across forest landscapes: A browsing extension for LANDIS-II. *Ecological Modelling*, 350, 11–29.
- Early R, Keith SA. 2019. Geographically variable biotic interactions and implications for species ranges. *Glob Ecol Biogeogr* 28(1):42–53. <https://doi.org/10.1111/geb.12861>.
- Evans P, Brown CD. 2017. The boreal-temperate forest ecotone response to climate change. *Environ Rev* 25(4):423–431. <https://doi.org/10.1139/er-2017-0009>.
- Evans P, Crofts AL, Brown CD. 2020. Biotic filtering of northern temperate tree seedling emergence in beyond-range field experiments. *Ecosphere* 11(5):e03108. <https://doi.org/10.1002/ecs2.3108>.
- Fahey, T. J., Cleavitt, N. L., Templer, P. H., Groffman, P. M., Bailey, A. S., Caron, S. B., & Wilson, G. (2025). Foliar resorption of beech and maple along an elevation gradient in a northern hardwood forest. *Forest Ecosystems*, 13, 100304.
- Fernando DR, Marshall AT, Lynch JP. 2016. Foliar nutrient distribution patterns in sympatric maple species reflect contrasting sensitivity to excess manganese. *PLoS One* 11(7):e0157702. <https://doi.org/10.1371/journal.pone.0157702>.
- Fischelli N, Frelich LE, Reich PB. 2012. Sapling growth responses to warmer temperatures 'cooled' by browse pressure. *Glob Change Biol* 18(11):3455–3463. <https://doi.org/10.1111/j.1365-2486.2012.02785.x>.
- Frelich LE, Peterson RO, Dovčiak M, Reich PB, Vucetich JA, Eisenhauer N. 2012. Trophic cascades, invasive species and body-size hierarchies interactively modulate climate change responses of ecotonal temperate–boreal forest. *Philos Trans R Soc B: Biol Sci* 367(1605):2955–2961. <https://doi.org/10.1098/rstb.2012.0235>.

- Foster JR, D'Amato AW. 2015. Montane forest ecotones moved downslope in northeastern USA in spite of warming between 1984 and 2011. *Glob Change Biol* 21(12):4497–4507. <https://doi.org/10.1111/gcb.13046>.
- Galen C. 1990. Limits to the distributions of alpine tundra plants: herbivores and the alpine skypilot, *Polemonium viscosum*. *Oikos*. <https://doi.org/10.2307/3545146>.
- Gilliam FS, Burns DA, Driscoll CT, Frey SD, Lovett GM, Watmough SA. 2019. Decreased atmospheric nitrogen deposition in eastern North America: predicted responses of forest ecosystems. *Environ Pollut* 244:560–574.
- Gilliam FS, Burns DA, Driscoll CT, Frey SD, Lovett GM, Watmough SA. 2024. Responses of forest ecosystems to decreasing nitrogen deposition in eastern North America. In: *Atmospheric nitrogen deposition to global forests* (pp. 205–225). Academic Press.
- Goodwin K. 2025. Herbivory and climate as drivers of plant population and range dynamics (Doctoral dissertation, University of British Columbia).
- Halman JM, Schaberg PG, Hawley GJ, Hansen CF, Fahey TJ. 2015. Differential impacts of calcium and aluminum treatments on sugar maple and American beech growth dynamics. *Can J For Res* 45(1):52–59. <https://doi.org/10.1139/cjfr-2014-0250>.
- Hane EN, Hamburg SP, Barber AL, Plaut JA. 2003. Phytotoxicity of American beech leaf leachate to sugar maple seedlings in a greenhouse experiment. *Can J For Res* 33(5):814–821. <https://doi.org/10.1139/x03-028>.
- Hargreaves AL. 2024. Geographic gradients in species interactions: from latitudinal patterns to ecological mechanisms. *Ann Rev Ecol Syst* 55:369–393.
- HilleRisLambers J, Harsch MA, Ettinger AK, Ford KR, Theobald EJ. 2013. How will biotic interactions influence climate change-induced range shifts? *Ann N Y Acad Sci* 1297(1):112–125. <https://doi.org/10.1111/nyas.12182>.
- Houle D, Renaudin M, Duchesne L, Moore JD, Benoist A. 2024. Soil solution chemistry weak response to long-term N addition points towards a strong resilience of northeastern American forests to past and future N deposition. *Sci Total Environ* 946:174387.
- Huang CYL, Schulte EE. 1985. Digestion of plant tissue for analysis by ICP emission spectroscopy. *Commun Soil Sci Plant Anal* 16(9):943–958. <https://doi.org/10.1080/00103628509367657>.
- Janowiak MK, D'Amato AW, Swanston CW, Iverson L, Thompson FR, Diak WD, Matthews S, Peters MP, Prasad A, Fraser JS, Brandt LA, Butler-Leopold P, Handler SD, Shannon PD, Burbank D, Campbell J, Cogbill C, Duveneck MJ, Emery MR, Templer PH. 2018. New England and Northern New York Forest Ecosystem Vulnerability Assessment and Synthesis: A Report from the New England Climate Change Response Framework Project, 234.
- Janzen DH. 1970. Herbivores and the number of tree species in tropical forests. *Am Nat* 104(940):501–528.
- Koitzsch KB. 2002. Application of a moose habitat suitability index model to Vermont wildlife management units. *Alces* 38:89–107.
- Lawlor JA, Comte L, Grenouillet G, Lenoir J, Baecher JA, Bandara RMWJ, Sunday J. 2024. Mechanisms, detection and impacts of species redistributions under climate change. *Nat Rev Earth Environ* 5(5):351–368.
- Lawrence GB, Hazlett PW, Fernandez JJ, Ouimet R, Bailey SW, Shortle WC, Smith KT, Antidormi MR. 2015. Declining acidic deposition begins reversal of forest-soil acidification in the northeastern U.S. and eastern Canada. *Environ Sci Technol* 49(22):13103–13111. <https://doi.org/10.1021/acs.est.5b02904>.
- Leck MA, Parker VT, Simpson R, Eds. 2008. *Seedling ecology and evolution*. Cambridge: Cambridge University Press.
- Maron JL, Crone E. 2006. Herbivory: effects on plant abundance, distribution and population growth. *Proc R Soc B Biol Sci* 273(1601):2575–2584. <https://doi.org/10.1098/rspb.2006.3587>.
- Maron JL, Baer KC, Angert AL. 2014. Disentangling the drivers of context-dependent plant–animal interactions. *J Ecol* 102(6):1485–1496. <https://doi.org/10.1111/1365-2745.12305>.
- McIntosh JL. 1969. Bray and Morgan soil extractants modified for testing acid soils from different parent materials. *Agron J* 61(2):259–265.
- Ni M, Vellend M. 2024. Soil properties constrain predicted poleward migration of plants under climate change. *New Phytologist* 241(1):131–141. <https://doi.org/10.1111/nph.19164>.
- Nyland RD, Bashant AL, Bohn KK, Verostek JM. 2006. Interference to hardwood regeneration in northeastern North America: Ecological characteristics of American beech, striped maple, and hobblesh. *North J Appl For* 23(1):53–61. <https://doi.org/10.1093/njaf/23.1.53>.
- Phelan JN, Van Houtven G, Clark CM, Buckley J, Cajka J, Hargrave A, Sabo RD. 2024. Climate change could negate US forest ecosystem service benefits gained through reductions in nitrogen and sulfur deposition. *Sci Rep* 14(1):10767.
- R Development Core Team. 2024. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. ISBN 3-900051-07-0, URL: <http://www.R-project.org>.
- Rammer W, Thom D, Baumann M, Braziunas K, Dollinger C, Kerber J, Seidl R. 2024. The individual-based forest landscape and disturbance model iLand: Overview, progress, and outlook. *Ecol Model* 495:110785.
- Rivest S, Vellend M. 2018. Herbivory and pollen limitation at the upper elevational range limit of two forest understory plants of eastern North America. *Ecol Evol* 8(2):892–903. <https://doi.org/10.1002/ece3.3397>.
- Schaberg PG, Tilley JW, Hawley GJ, DeHayes DH, Bailey SW. 2006. Associations of calcium and aluminum with the growth and health of sugar maple trees in Vermont. *For Ecol Manag* 223(1–3):159–169. <https://doi.org/10.1016/j.foreco.2005.10.067>.
- Siccama TG. 1974. Vegetation, soil, and climate on the green mountains of Vermont. *Ecol Monogr* 44(3):325–349. <https://doi.org/10.2307/2937033>.
- Sullivan TJ, Lawrence GB, Bailey SW, McDonnell TC, Beier CM, Weathers KC, McPherson GT, Bishop DA. 2013. Effects of acidic deposition and soil acidification on sugar maple trees in the Adirondack Mountains, New York. *Environ Sci Technol* 47(22):12687–12694. <https://doi.org/10.1021/es401864w>.
- Thom D, Rammer W, Albrich K, Braziunas KH, Dobor L, Dollinger C, Seidl R. 2024. Parameters of 150 temperate and boreal tree species and provenances for an individual-based forest landscape and disturbance model. *Data Brief* 55:110662.

- Tourville JC, Wason JW, Dovciak M. 2022. Canopy gaps facilitate upslope shifts in montane conifers but not in temperate deciduous trees in the Northeastern United States. *J Ecol* 110(12):2870–2882. <https://doi.org/10.1111/1365-2745.13993>.
- Tourville JC, Zarfos MR, Lawrence GB, McDonnell TC, Sullivan TJ, Dovciak M. 2023. Soil biotic and abiotic thresholds in sugar maple and American beech seedling establishment in forests of the northeastern United States. *Plant Soil* 491:387–400. <https://doi.org/10.1007/s11104-023-06123-2>.
- Tourville J, Dovciak M. 2023. Small mammal granivory as a biotic filter for tree establishment beyond elevation range boundaries. *Plant Ecol* 224(5):435–447. <https://doi.org/10.1007/s11258-023-01305-5>.
- Tourville JC, Horton TR, Dovciak M. 2024. Mycorrhizal fungi as critical biotic filters for tree seedling establishment during species range expansions. *Ecol Monogr*. <https://doi.org/10.1002/ecm.1634>.
- Urli M, Brown CD, Perez RN, Chagnon PL, Vellend M. 2016. Increased seedling establishment via enemy release at the upper elevational range limit of sugar maple. *Ecology* 97(11):3058–3069. <https://doi.org/10.1002/ecy.1566>.
- Van der Putten WH. 2012. Climate change, aboveground-belowground interactions, and species' range shifts. *Ann Rev Ecol Evol Syst* 43(1):365–383. <https://doi.org/10.1146/annurev-ecolsys-110411-160423>.
- Vellend M, B     M, Carteron A, Crofts AL, Danneyrolles V, Gamhewa HT, Watts DA. 2021. Plant responses to climate change and an elevational gradient in Mont M  gantic National Park, Qu  bec, Canada. *Northeast Nat* 28(sp11):4–28. <https://doi.org/10.1656/045.028.s1102>.
- Walck JL, Hidayati SN, Dixon KW, Thompson KEN, Poschlod P. 2011. Climate change and plant regeneration from seed. *Glob Change Biol* 17(6):2145–2161. <https://doi.org/10.1111/j.1365-2486.2010.02368.x>.
- Wason JW, Dovciak M. 2017. Tree demography suggests multiple directions and drivers for species range shifts in mountains of northeastern United States. *Glob Change Biol* 23(8):3335–3347. <https://doi.org/10.1111/gcb.13584>.
- Wetzel WC, Inouye BD, Hahn PG, Whitehead SR, Underwood N. 2023. Variability in plant–herbivore interactions. *Ann Rev Ecol Evol Syst* 54(1):451–474. <https://doi.org/10.1146/annurev-ecolsys-102221-045015>.
- Williams CE, Mosbacher EV, Moriarty WJ. 2000. Use of turtlehead (*Chelone glabra* L.) and other herbaceous plants to assess intensity of white-tailed deer browsing on Allegheny Plateau riparian forests, USA. *Biol Conserv* 92(2):207–215. <https://doi.org/10.1111/j.1752-4571.2007.00013.x>.
- Wills A, Beier CM, Lawrence GB, Drake JE. 2023. Foliar chemical composition and respiration rates of sugar maple (*Acer saccharum*) and American beech (*Fagus grandifolia*) trees across a gradient of soil acidification. *Biogeochemistry* 162(3):327–341. <https://doi.org/10.1007/s10533-022-01010-x>.
- Zarfos MR, Dovciak M, Lawrence GB, McDonnell TC, Sullivan TJ. 2019. Plant richness and composition in hardwood forest understories vary along an acidic deposition and soil-chemical gradient in the northeastern United States. *Plant Soil* 438:461–477. <https://doi.org/10.1007/s11104-019-04031-y>.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.